Yingyu Wang

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Experience

The University of Edinburgh — Edinburgh, UK
Research Associate, School of Informatics — Jun. 2026 – Present

- Supervisor: A/Prof. Liang Zhao

University of Technology Sydney — Sydney, Australia
Postdoctoral Research Associate, School of Mechanical and Mechatronic Engineering — Sep. 2025 – May 2026

- Supervisor: Prof. Shoudong Huang

The University of Edinburgh — Edinburgh, UK
Visiting Ph.D., School of Informatics — Mar. 2025 – Sep. 2025

- Topic: High-Fidelity 3D LiDAR Mapping for Autonomous Driving

University of Technology Sydney — Sydney, Australia
Teaching Assistant, School of Mechanical and Mechatronic Engineering — Aug. 2022 – Dec. 2025

- Design Optimisation for Manufacturing (UTS 49928)
- Robotics Studio 1 (UTS 41068)

Education

University of Technology Sydney — Sydney, Australia
Ph.D. in Robotics — Jul. 2021 – Dec. 2025

- Supervisor: Prof. Shoudong Huang and A/Prof. Liang Zhao
- Thesis Title: *Novel Non-feature Based SLAM: Joint Optimization of Non-feature Maps and Robot Trajectories*

Ocean University of China — Qingdao, China
Master in Computer Science — Sep. 2017 – Jun. 2020

- Supervisor: Prof. Junyu Dong
- Thesis Title: *Research and Application of Depth Estimation and Semantic Segmentation Algorithms for Agricultural UAVs*

Chengdu University of Technology — Chengdu, China
Bachelor in Computer Science — Sep. 2013 – Jun. 2017

Publications

[1] **Y. Wang**, L. Zhao, and S. Huang, "Occupancy-SLAM: An Efficient and Robust Algorithm for Simultaneously Optimizing Robot Poses and Occupancy Map", *IEEE Transactions on Robotics (T-RO)*, vol. 41, pp. 4057–4077, 2025.

[2] **Y. Wang**, L. Zhao, and S. Huang, "Grid-based Submap Joining: An Efficient Algorithm for Simultaneously Optimizing Global Occupancy Map and Local Submap Frames", in *2024 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, IEEE, 2024, pp. 10 121–10 128.

[3] L. Zhao, **Y. Wang**, and S. Huang, "Occupancy-SLAM: Simultaneously Optimizing Robot Poses and Continuous Occupancy Map", in *Proceedings of Robotics: Science and Systems (RSS)*, New York City, NY, USA, Jun. 2022.

[4] Y. Zhou, **Y. Wang**, L. Zhao, and S. Huang, "Correspondence-free Multiview Point Cloud Registration via Depth-Guided Joint Optimisation", in *2025 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, IEEE, 2025, pp. 10 121–10 128.

[5] **Y. Wang**, Y. Ju, M. Jian, K.-M. Lam, L. Qia, and J. Dong, "Self-supervised depth completion with attention-based loss", in *International Workshop on Advanced Imaging Technology (IWAIT) 2020*, SPIE, vol. 11515, 2020, pp. 517–524.

[6] Y. Zhang, **Y. Wang**, J. Dong, L. Qi, H. Fan, X. Dong, M. Jian, and H. Yu, "A joint guidance-enhanced perceptual encoder and atrous separable pyramid-convolutions for image inpainting", *Neurocomputing*, vol. 396, pp. 1–12, 2020.

[7] Y. Ju, X. Dong, **Y. Wang**, L. Qi, and J. Dong, "A dual-cue network for multispectral photometric stereo", *Pattern Recognition*, vol. 100, p. 107 162, 2020.

[8] Y. Ju, M. Jian, S. Guo, **Y. Wang**, H. Zhou, and J. Dong, "Incorporating lambertian priors into surface normals measurement", *IEEE Transactions on Instrumentation and Measurement*, vol. 70, pp. 1–13, 2021.

[9] J. Li, H. Ren, J. Song, **Y. Wang**, Y. Zhang, H. Fan, and J. Dong, "UMB-GSfM: Underwater marker-based global structure-from-motion", in *2025 IEEE International Conference on Multimedia and Expo Workshops (ICMEW)*, IEEE, 2025, pp. 1–6.

[10] Y. Wei, H. Bi, Y. Liu, J. Sun, **Y. Wang**, L. Qi, J. Dong, and H. Fan, "Stereo vision SLAM for real-time underwater dense 3D mapping", in *2025 IEEE International Conference on Multimedia and Expo Workshops (ICMEW)*, IEEE, 2025, pp. 1–6.

Patent

1. Junyu Dong, Jie Gao, Hao Fan, Jun Chi, **Yingyu Wang**, 3D Imaging Device and Method based on Multi-spectral Photometric Stereo and Laser Scanning, *Invention Patent in China.*

Professional Service

- Associate Editor for IROS in 2026
- Reviewer
  Journal: IJRR, T-RO, R-AL, AURO
  Conference: RSS in 2025, ICRA in 2024–2026, IROS in 2023–2025
- Other Service
  Lead Organizer, RSS 2026 Workshop: *From Perception to Action: Representation-Centric Robot Autonomy*
  Assisted Prof. Shoudong Huang with RSS 2023 publication duties and RSS 2024 registration duties
  Assisted Prof. Shoudong Huang with the IEEE RAS Winter School on SLAM in Deformable Environments, 2021

Projects

- Large-Scale Graph Optimisation
  (Postdoctoral Research Project at The University of Edinburgh)
  This project investigates scalable optimisation methods for large graph-structured problems, focusing on computational efficiency, memory scalability, and solution quality. The work develops algorithmic and implementation strategies for high-dimensional sparse optimisation problems arising in robotics and related estimation tasks.

- Novel Non-feature Based SLAM: Joint Optimization of Non-feature Maps and Robot Trajectories
  (Ph.D. Thesis at University of Technology Sydney)
  A novel approach for jointly optimizing robot trajectories and non-feature maps (occupancy grid maps and signed distance function maps) to obtain more accurate estimates of both poses and the environment. The project consists of two main components: joint optimization of poses and non-feature maps and novel submap joining. Until now, there are 3 published papers related to this project. The implementation is carried out in C++ and MATLAB.

- High-Fidelity 3D LiDAR Mapping for Autonomous Driving
  (Cooperated with Robust Autonomy and Decisions Group at The University of Edinburgh)
  This project focuses on generating high-fidelity 3D point clouds using LiDAR sensing for road sign and lane line recognition in autonomous driving scenarios. It also aims to enable robust and accurate localization in environments where GPS/GNSS signals are unreliable or unavailable, such as tunnels, bridges, and overpasses.